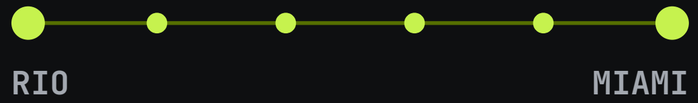
TESTNET · TEST KEYS · NOT REAL CUSTODIANS

Radox Custody Rail

Know who holds the goods: every handover signed by both custodians.

A dual-signed custody record on Stellar · First corridor: Rio–Miami

Scan: the live shipment
rio-miami-rail.vercel.app



Hack Meridian / 2026 LISBON | GUAVA® Stellar



© 2026 HackMeridian · Lisbon

PMF · THE PROBLEM

Who holds the cargo right now?

Between seller and buyer, a shipment changes hands several times.



Handovers are mostly paper, or sit inside one operator's own system.

US$2.5T

Global trade finance gap.

ADB Global Trade Finance Gap Survey, Dec 2025. It sizes the financing gap; it is not evidence of the custody problem.

Our hypothesis: a financier cannot see who holds the goods before paying. No customer has confirmed it yet; we are testing it with mentors.

Context: our explainer video about a Rio–Miami trade-corridor initiative poses one possibility: a unique digital record for every shipment that both sides can check. This project is a custody layer for that record.

© 2026 HackMeridian · Lisbon · #HackMeridian26

THE SOLUTION

Both custodians sign, or nothing changes

Outgoing custodian
`signs`

One Stellar transaction
`custody_registry.handover`

Incoming custodian
`signs`

Either signature missing → the call fails.

— Each handover moves custody and extends an on-chain hash chain, the `head`.

— The token stays with its holder in transit; only custody changes.

— After Delivered, the operator burns the token with the existing `redeem`[1].

A custody record keyed to the container number. In production it would sit beside the carrier's bill of lading; not integrated in this build.

[1] Radox platform policy, not enforced on-chain by the registry; redeem still needs the holder's signature.

© 2026 HackMeridian · Lisbon · #HackMeridian26

THE PRODUCT · LIVE ON TESTNET

Anyone can check the chain. No login.

Warehouse → Port, both signed	ledger 5381142 · 635c99…f0bd38
Port → Carrier, both signed	ledger 5381143 · 56a0f0…7b7cc2
Event attested by Source	ledger 5381144 · 333792…068905
Carrier → Customs, both signed	ledger 5381145 · ac268a…c3c4b2
Customs → Recipient, both signed	ledger 5381146 · 5199d4…400b7a
Delivered by Recipient to Holder, both signed	ledger 5381147 · 919def…144e37
Redeemed: token burned	864fd1…15644d

Verify head

Verify head recomputes the hash chain in the browser and compares it with the head stored in the contract.

MATCH

Clicked live on 25 Oct 2026.

Read-only: Stellar RPC and Horizon, no wallet.

rio-miami-rail.vercel.app

© 2026 HackMeridian · Lisbon · #HackMeridian26

TECHNICAL WORKFLOW

7 testnet transactions. All successful.

open	handover 1	handover 2	handover 3	handover 4	deliver	burn
operator	Warehouse → Port	Port → Carrier	Carrier → Customs	Customs → Recipient	Recipient → Holder	existing redeem
2b3d51f0	635c99ac	56a0f006	ac268a49	5199d4fb	919def8c	864fd164

7
successful testnet transactions

9
invalid calls refused (simulated)

24
contract unit tests pass

MATCH
verify-head rebuilds the chain

`custody_registry` CCBJVXMJYWPHTDJHDFPRGKPGXGEP24JOPJAMUS5C3BMZJTRTWR2H4LII

Plus one event attestation by a source custodian (33379211), also on chain. Full hashes: the repo's proofs/lifecycle.json.

Every actor is a test key held by one harness: this proves the mechanism, not independent custodians.

© 2026 HackMeridian · Lisbon · #HackMeridian26

ECOSYSTEM · WHY STELLAR

Why Stellar: two signers, one call

01	**Multi-party auth in one call**	Soroban authorises both custodians inside one contract invocation: the handover is atomic.
02	**Events plus an on-chain head**	RPC keeps contract events about 7 days. The head stored in the contract lets anyone rebuild the full history later.
03	**The token stays put, then burns**	The shipment token never moves in transit; the existing Radox redeem burns it after delivery.

Nearest neighbours we found (LumenLoop, SCF, Scout; searched 24–25 Oct 2026): Kutana and Trustless Work release escrow on trade milestones. We found none that records a custody handover signed by two custodians, and no SEP for it.

Open source, Apache-2.0: a custody primitive other Stellar real-world-asset issuers can reuse.

© 2026 HackMeridian · Lisbon · #HackMeridian26

USERS · BUSINESS MODEL

Custodians sign. Financiers check.

CUSTODIANS	Warehouse, port, carrier, customs: sign each handover.
EXPORTER · BUYER	See where the goods are and who holds them.
FINANCIER	Checks the custody chain before paying.
OPERATOR	A financial institution, the client, would run the rail for its trade flows.

MODEL · THE OPERATOR PAYS

SWIFT-style: the record format is public; Radox runs the rail and the API.

Revenue: a fee from the financial institution that operates the rail. Not priced yet.

Radox is the rail, never the issuer. Regulation and licences stay with the client and its local counsel.

© 2026 HackMeridian · Lisbon · #HackMeridian26

HACKATHON PROGRESS

Built this weekend, on disclosed prior work

PRIOR WORK · BEFORE SUN 25 OCT 09:00

— Radox platform and 3 Soroban contracts on Stellar mainnet (public repo, created 2026-08-19)

— transfer_agent and its redeem (new testnet instance from the existing wasm)

— Planning files, 24–25 Oct

BUILT AT HACKMERIDIAN

— custody_registry contract, 24 unit tests

— Lifecycle harness: 7 testnet transactions, 9 refused calls

— verify-head script and the public shipment page

— Radox API reads the registry: same head (closed backend, testnet)

AI attribution Code, tests and scripts were written with AI coding agents (Claude Code); the team decided scope and what ships. Per-file record in AI_USAGE.md. The video narration is an AI voice generated from Pedro's voice.

© 2026 HackMeridian · Lisbon · #HackMeridian26

NEXT · THE ASK

Next: one pilot shipment, not yet agreed

NOW

Testnet this weekend: contract, 7 transactions, public page, API read.

NEXT

Full testnet test matrix. A first pilot shipment with one producer (none agreed yet).

PLANNED

SCF Build application (not yet submitted). Security audit before mainnet (not done).

FUTURE

Mainnet. A USD on-ramp leg (not built).

No users or revenue yet. First users come from one pilot shipment in the Rio–Miami corridor.

Our ask **Introductions to one financial institution and one logistics operator for a first pilot shipment, and a Stellar engineer's review of the design.**

© 2026 HackMeridian · Lisbon · #HackMeridian26

THE TEAM

Why us: a founder already on Stellar mainnet

Pedro Wakigawa Saragossy

Founder, captain. Built Radox: 3 Soroban contracts on Stellar mainnet.

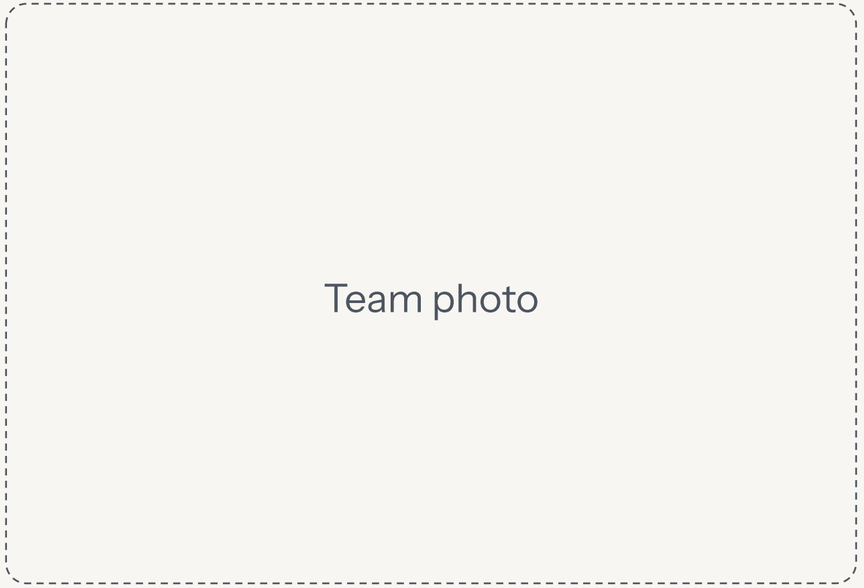
Leo Schlanger

Developer: custody_registry, the testnet harness and the proofs.

Vhendala

QA and product review: reads every deliverable as a judge would.

Leo and Vhendala joined for the hackathon.



Live page rio-miami-rail.vercel.app **Code** github.com/pwsaragossy/rio-miami-rail (Apache-2.0)

AI Code written with AI coding agents (Claude Code). Narration: an AI voice from Pedro's voice. No AI-generated images in this deck.

© 2026 HackMeridian · Lisbon · #HackMeridian26